The empirical recurrence for OEIS A267461, proved from an exact model

Adrian Perez Fontelles
Independent researcher

12 September 2026

Abstract

OEIS A267461 carries an empirical recurrence of order 4 contributed by Colin Barker. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a certified block template for the automaton's row, from which a provably satisfies a MONIC linear recurrence of order $S = 12$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

2020 Mathematics Subject Classification. 68Q80, 11B85, 05A15.

1 The conjecture

OEIS A267461 is "Total number of OFF (white) cells after n iterations of the "Rule 133" elementary cellular automaton starting with a single ON (black) cell.". It has offset 0 and begins

$$0, 2, 6, 12, 18, 26, 34, 44, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Conjectures from _Colin Barker_, Jan 16 2016: (Start) a(n) = 2*a(n-1) - 2*a(n-3) + a(n-4) for n>4.

As of the "Last modified" line on the live entry (Feb 16 2025 08:33:29, revision 19) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The entry counts the running total of its OFF (white) cells over the first n iterations of the automaton. Nothing is being enumerated over a window, so there is no digraph and no transfer matrix. What settles the sequence is the shape of the row itself: the row is a concatenation of a bounded number of blocks, of which some are FIXED and some are REPEATED a number of times growing by one every p steps. Writing r for the residue of n modulo the period p and j for the number of periods past a pre-period n_0, the automaton was run and

$$w(n_0 + r + jp) \ = \ B_0\, Q_1^j\, B_1 \cdots Q_m^j\, B_m$$

was verified for every available n, the blocks depending only on r. The row at stage n is the light cone, of width $2n+1$, so $\sum_i |Q_i| = 2p$, and counting ON cells along a residue class gives

$$\mathrm{on}(n_0 + r + jp) \ = \ \sum_i \mathrm{ones}(B_i) \ + \ j \sum_i \mathrm{ones}(Q_i),$$

which is LINEAR in j.

The template is a theorem and not a fit. The rule is a local map of radius 1, so p steps have radius p. Let the template hold at $j = K$ with every repeated run longer than $4p+4|Q_i|$, and let $F^p(T(K)) = T(K+1)$ and $F^p(T(K+1)) = T(K+2)$ be verified by direct simulation — which they were. For $j > K$, $T(j)$ is $T(K)$ with further copies of each Q_i inserted inside runs already longer than $2p$. An output cell within distance $|B_0| + |Q_1|K - p$ of the left end sees only input cells that $T(j)$ and $T(K)$ share at the same left-distance, so it agrees with the corresponding cell of $F^p(T(K))$; symmetrically at the right end; and strictly inside a run of length exceeding $2p$ the input is $|Q_i|$-periodic, so the output there is $|Q_i|$-periodic with the period and phase already fixed by the two verified steps. Hence $F^p(T(j)) = T(j+1)$ for every $j \geq K$, and two simulations prove infinitely many.

For this entry the automaton is rule 133, the period is $p = 2$ and the pre-period is $n_0 = 2$. The certificate is

$$r = 0: \quad \mathtt{0} \cdot \mathtt{0101}^j \cdot \mathtt{0100}$$
$$r = 1: \quad \mathtt{00} \cdot \mathtt{0101}^j \cdot \mathtt{01000}$$

3 The bound

So on is quasi-linear in n with period p and is annihilated by $(z^p - 1)^2$. The row has width $2n + 1$, so $\mathrm{off}(n) = 2n + 1 - \mathrm{on}(n)$ is quasi-linear too and $(z - 1)^2$ divides $(z^p - 1)^2$, giving it the same annihilator; a running total is a partial sum and contributes one further factor $(z-1)$. The pre-period raises the numerator's degree by at most n_0, so the recurrence of order $2p+1$ is in force from index $n_0 + 2p + 1$ and the order $S = 12$ used below covers all four wordings.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^4 - \sum_i c_i z^{4-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. □

Theorem 2.

$$a(n) \;=\; 2\,a(n-1) - 2\,a(n-3) + a(n-4)$$

for every $n > 4$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 4+1$ onwards, and the run exceeds $S+4$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 4$ the recurrence fails on the entry's own published terms, so no larger range holds. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 52 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A267461.

[2] M. Beck and S. Robins, *Computing the Continuous Discretely*, 2nd ed., Springer, 2015. (Ehrhart quasi-polynomials and their periods.)

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.

[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.